The Effect of Female Leadership on the Wage-Productivity Gap: Evidence from Colombian Manufacturing Firm Data

Applied Microeconomic Theory ECON 6301 Paper

Ingri Katherine Quevedo Rocha

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ingriq@gwu.edu
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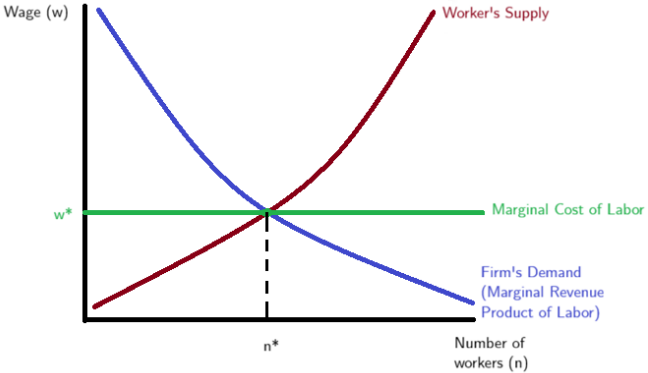
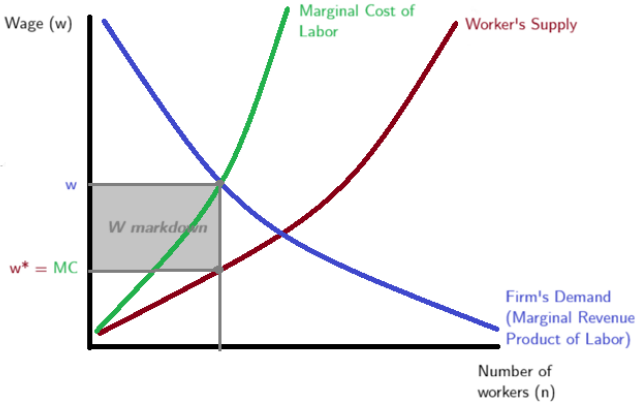
Microeconomic Theory: How Wages Are Set in an Ideal World

- Many firms compete to hire workers
- Workers can easily switch jobs
- Wages reflect how much workers contribute to production



Microeconomic Theory: When Firms Have Power Over Workers

- Fewer job options for workers
- Switching jobs is costly or risky
- Firms can influence wages



Research Question: Do women-led firms have lower labor markdowns?

$$Markdown_{it} = \beta_0 + \beta_1 ShareFemaleOwners_{it} + \Phi Controls_{it} + \epsilon_{it} \quad (1)$$

Expected $\beta 1 < 0$

- Management choices shape wage-setting and competition Bertrand and Mullainathan (1998)
- Firms differ systematically in management practices Bloom and Van Reenen (2007)
- Labor market power affects women more strongly (Hirsch et al. (2010), Sharma (2023))
- Female leaders reduce gender-specific wage distortions (Flabbi et al. (2017), Flabbi et al. (2019))
- Firms with female leadership are less likely to downsize and pursue value-destroying consolidation Matsa and Miller (2013), Levi et al. (2014), Matsa and Miller (2014)

Brief Literature Review

1. **Labor market power and markdowns**

   - Labor market power is widespread: it can arise from employer concentration, job amenities, or information/search frictions (Manning (2021); Card (2022); Azar et al. (2019); Manning (2013); Lentz and Mortensen (2010); Moscarini and Postel-Vinay (2010)), or higher concentration to lower wages (Azar et al. (2017); Azar et al. (2019); Benmelech et al. (2018); Rinz (2022)).

2. **Latin America / Colombia evidence**

   - In Latin America and the Caribbean workers receive about half of what they contribute on average. In contrast, workers in the U.S. and advanced economies receive 65 percent and 81 percent, respectively.
   - Colombia: estimated firm-level labor supply elasticity ≈ 2.5, implying workers produce about 40% more than they earn (Amodio and de Roux (2024)).

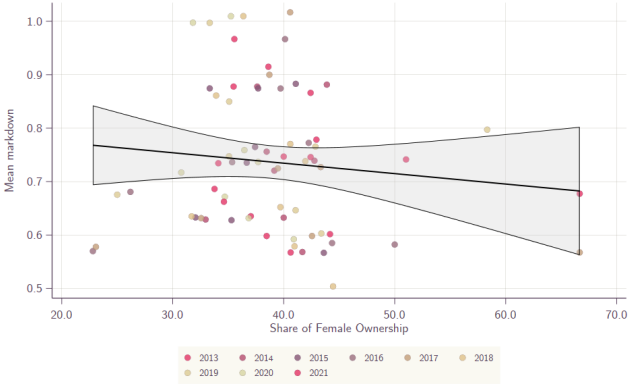
Brief Literature Review

3. Gender, leadership, and firm behavior

- Female leadership correlates with firm performance in Latin America Flabbi et al. (2017) and with improved productivity and employment stability in U.S. firms (Tunyi et al. (2023), Matsa and Miller (2014)).
- Gender differences in labor market power matter: in Chile, differences in labor supply elasticities imply men would earn about 22% more than women Sánchez et al. (2022)
- Female executives may reduce gender-specific wage distortions by better interpreting productivity signals Flabbi et al. (2019), and women/minority workers can be more exploitable due to historical and social constraints Stelzner and Bahn (2022).

Data and Correlation Evidence

- Colombia Manufacturing Survey (firm-year, 1998-2024) **Issue:** Markdown estimation is complex and time consuming
- IADB CompeteLAC Markdowns (industry-year, 1998-2021) **Issue:** Only 20 industries by year



Source: Author's calculation using EAM and CompeteLAC.
Note: Markdown with Labor as input, Firm Wage Bill-weighted, method Translog Adjusted

Preliminary findings: Negative but not significant correlation between industry-level markdowns and female ownership share.

Conclusion and Open Questions

This paper examines how female ownership relates to labor markdowns. It highlights that firms often exert labor market power rather than behaving as pure wage takers.

Open questions:

- Do results change when moving from industry-level to firm-level data?
- Is the negative relationship driven by higher productivity in female-led firms rather than wage setting alone?
  - In Colombia female leadership is associated with higher firm performance (Moreno-Gómez and Calleja-Blanco (2018)). Larger, more productive firms pay higher wages, yet relative to workers' contributions to the value of the firm, the ratio might be larger, leading to higher markdowns.
- Are effects stronger in more concentrated or friction-intensive labor markets? I would include industry HHI in the regression.

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